



RESEARCH AND PROFESSIONAL DEVELOPMENT PLANNING

Version 1.3.1



UCD Graduate Studies





Introduction

Research and Professional Development planning (RPDP) is an integral part of the Structured PhD programme at UCD. Its purpose is to ensure that your work is clearly focused on achieving your research and professional goals. This will play a major part in informing the trajectory of your PhD research and in your training and development as a researcher. Your RPDP will help you develop key skills that will be invaluable for both your current research and your future career prospects.

The Structured PhD requires a minimum of 30 taught credits that you must document in the RPDP. These credits are central to the PhD experience and fundamental to acquiring transferable skills. Research Masters students may take taught modules relevant to their area of research or professional development as required or recommended by their Principal Supervisor.

In line with University policy, PhD candidates can apply for recognition of prior/concurrent, learning which will not normally amount to more than 30% of the total requirement.

A further mandatory part of your doctoral programme will be the satisfactory completion of Research Integrity Training. This can take the form of an online module or a blended module (SCI50020) that you must record in your RPDP.

The RPDP will also inform discussions at your meetings with your supervisor and Research Studies Panel (RSP). You will find a template in this document which you can use to structure the meetings.

For PhD students, the RPDP record will be a mandatory input to your Stage Transfer Assessment which will determine whether you progress from Stage 1 to Stage 2 of your doctoral studies. The Stage Transfer Assessment Panel will normally consist of three members of the permanent faculty of the University. Your Principal/ co-supervisor will not be a member of the panel. The Stage Transfer Assessment will normally occur within 5 trimesters of registration for a full-time student and 7 trimesters for part-time.

Your RPDP form is treated as confidential and is normally only available to your supervisor and RSP. In addition, as part of the Stage 1 to Stage 2 assessment process the Transfer Assessment Panel, staff associated with the Transfer Assessment process (the School Administrator, the Governing Board) will also have access to the RPDP.



There are three main components to your Research and Professional Development plan:

1. A RESEARCH PLAN

The purpose of your Research Plan is to provide you with a clear research focus and a coherent work programme. A well-structured Research Plan will enable you to review your progress and adjust your objectives as your research evolves.

Keeping your Research Plan up to date will help support your writing progress reports, preparing presentations and publications, drafting funding applications and completing your thesis.

2. A PROFESSIONAL DEVELOPMENT PLAN

On leaving UCD with a research degree it will be expected that, in addition to having produced a body of research, you will also have developed a set of generic and transferable skills that will enhance employability and career development.

UCD offers a range of opportunities to acquire transferable skills. We support a model of developing career skills in your research programme and from your research programme. Research degree holders are well-positioned in an increasingly cross-sectoral market that emphasises flexibility, creativity and critical thinking. [Research Student Training and Development](#)

3. RESEARCH STUDIES PANEL MEETING RECORD

Central to your programme are regular meetings with your supervisor and RSP to discuss your research, your professional development and progress in achieving your goals. A mandatory outcome of the meetings with your RSP is a formal record of your research and professional plans and your progress to date. These records will also inform your Stage Transfer Assessment.

The formal record form for these meetings is appended at the back of this document as a writeable PDF. This must be completed following each RSP meeting, signed by all members of your RSP, including yourself. Please [upload a signed copy to your SISWEB](#). You should also submit a signed copy to your School administrator.

- Your RSP will normally be comprised of your principal supervisor / co-supervisor and at least two (but not normally more than four) advisors, one of whom must be a UCD faculty member. The RSP will provide advice and support during the course of your doctoral studies and will assist in monitoring your progress. For more details about your RSP, please contact your School.



Research Planning

A course of study and research leading to the research master's degree is pursued in one stage. The primary purpose of master's research is to develop in the student the skills and competencies required to conduct effective research: it may include experiential research training and does not necessarily result in the creation of new knowledge. The distinguishing feature of doctoral research is that it is original research, the outcome of which is an original and substantial addition to knowledge and understanding.

A research plan is a structure which you design to organise your thoughts about your topic and schedule the specific tasks that you need to do, while building towards the successful completion of a structured programme of research.

Keep in mind that the objectives and goals you set yourself in your plan should be specific, challenging and achievable within a defined timeframe.

There is no such thing as an ideal research plan, but common to all is **structure** and **timeframe**: research is always planned in stages, each designed with some idea of how long it will take. There are two critical things to remember when you begin to design a research plan.

First, your supervisor is there to advise you. S/he will guide you as you formulate your plan. In consultation with your supervisor you should, within the first trimester of your PhD, be able to identify major tasks and work out the order in which they should be done. Members of your RSP will also discuss your plan at RSP meetings.

Second, research plans change as research progresses. The secrets of a successful research project are (a) the realisation that a research plan is always a work-in-progress and (b) the ability to adjust a project's structure and timeframe without losing sight of its goals.

At the start of your studies, your emphasis will be primarily on devising, developing and implementing your research. As you progress, your focus will also encompass the dissemination of your research results and the preparation of your thesis.

When developing your research plan, it may be helpful to consider the following:

ESSENTIAL ELEMENTS OF A RESEARCH PLAN

- Your research question or hypothesis and its importance
- Ensuring that all or any ethical requirements associated with your research are addressed
- Reviewing the relevant literature
- A plan of how you intend to investigate your hypothesis and interpret your results
- Reviewing your findings and incorporating any changes to the research plan
- Drawing conclusions
- How you intend to disseminate findings

DISSEMINATING YOUR RESEARCH

- Where do you intend to present the results of your work?
- Where do you intend to publish the results of your work?

WRITING UP YOUR THESIS

- Have you prepared a thesis plan?
- Are you familiar with how best to present your work in thesis form and what the requirements are?
- What progress have you made in writing chapters of your thesis?
- When do you hope to have your first draft completed?
- When do you hope to submit?

You will also need to set a timeframe to each of the aspects outlined above. Devising a realistic schedule for each step on the way is extremely important if you want to complete your PhD within the timeframe required.

The SMART guidelines indicate that objectives should be:



Specific	specific and clear statements about what is to be achieved.
Measurable	quantitative and/or qualitative (if objectives are not measurable then it is difficult to determine whether they have been achieved).
Achievable	objectives need to be realistic, challenging and motivating.
Relevant	relevant to the achievement of your research aims
Time bound	include a time-scale for in which to achieve your objective or goal.



Professional Development Planning

As a graduate research student, you will be supported by your supervisor / co-supervisor and RSP in identifying your educational training and personal and professional development needs. During the course of your research studies, you will have opportunities to acquire new skills and to hone existing ones.

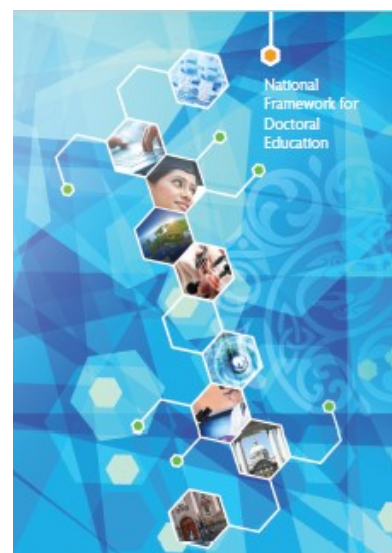
You should consult the [HEA's National Framework for Doctoral Education](#). It sets out the core principles of doctoral education in Ireland. In line with these principles, UCD strongly supports and encourages professional skills development. This includes opportunities to explore transferable skills training as well as potential employer networking and engagement. UCD's Careers Network offers a range of valuable support services, including a dedicated Career and Skills Consultant for graduate research students.

DEVELOPING THE RELEVANT SKILLS

First, identify the relevant skills you require for effective research and for your potential career paths. Every discipline area has its own particular suite of skills, and you may already have acquired many of these as a prerequisite to enrolling for a PhD programme. However, in order to develop career flexibility, it is advisable to acquire additional transferable and cross-sectoral skills.

Remember that you are encouraged to take modules and attend relevant workshops outside your School or College. You should consider identifying opportunities in the following broad areas: research skills; ethics and social understanding; communication and educational skills; personal effectiveness and development; team working and leadership; career management, in line with the [IUA Skills Statement](#).

For more information, please visit: [UCD Graduate Studies website](#).





RESEARCH AND PROFESSIONAL DEVELOPMENT FORM

Part 1 - RECORD OF STUDENT ACTIVITY

to be completed by the student

Candidate name and student number:

James Hannon 16430792

Date of meeting:

26/01/2023

Stage 1 or Stage 2 of the Study:

Stage 1

How many RSP meetings have you had this academic year:

0

Please state the dates of your previous RSP meetings:

29/06/2022, 30/11/2021

Section A

RECORD OF ONGOING RESEARCH

Please provide updates of your research progress (no more than 500 words):

I am currently researching the use of telematics data to assess the likelihood of an automobile insurance claim and the severity of such a claim. I have completed a literature review of 50 papers on this subject. Traditionally, the insurance industry has relied on a priori information to estimate the likelihood and severity of a claim. Telematics data offers a way to price this insurance dynamically, with premiums and risk changing with driver behaviours and tendencies.

I have also started researching the use of data from wearable devices to estimate the likelihood and severity of a health insurance claim. So far, I have reviewed 13 papers relating to this subject. This research is similar to the telematics research in that it intends to use novel sources of data to improve established methods.

As part of my individual project in trimester 2 of year 1 on this CRT I worked on statistical models for improved insurance pricing and risk assessment with my supervisor, Dr. O' Hagan. I used a publicly available dataset provided by Porto Seguro, which published a challenge of estimating claim likelihood for each policy holder. I used Gaussian mixture models to cluster observations and then performed logistic regression for the respective clusters. This experience was beneficial as I will be dealing with novel data that shares similar characteristics in my future research, such as data of mixed types (categorical, nominal, ordinal and numeric).

During my placement at the Central Bank of Ireland I worked on time dependent data, which we can expect to encounter when dealing with telematics or wearables data. I was able to use a similar approach of clustering, in the form of Dynamic Time Warping K-means, followed by supervised learning, which were LSTM and ARIMA models. Again, this experience was useful.



Section B

FUTURE PLAN

Please provide updates of your research plan for the following year (no more than 500 words):

Over the following year I plan on analysing telematics data provided by a global car rental company. This data set will contain information relating to vehicle maintenance, e.g. battery, oil, and fuel, as well as location information. It will be unstandardised and have no loss associated with it, making data cleaning important and an unsupervised learning approach necessary for analysis. Fortunately, I have experience with this type of data from my individual project and I will leverage my insights to tailor my approach.

I also plan to analyse wearables data from the UK Biobank study. This longitudinal study has been running since 2006. My supervisor and I have been in contact with Dr. Mark Farrell from Queen ' s University with aims of collaborating on a paper. Dr. Farrell has previously published a paper, " A Conceptual Model for Pricing Health and Life Insurance Using Wearable Technology " , which details how data from wearables can be incorporated into a health score for life insurance pricing models. An application to the UK Biobank exists titled, " Using wearable and health data to model and classify individual health risk " . We will be added to this existing application as collaborators to access the data.

The telematics and wearables datasets which we will encounter contain a high number of dimensions and may have time dependency. The dimensionality will have to be reduced to deploy many statistical techniques, e.g. regression. This can be achieved by clustering or principal component analysis, which I used in my group project in trimester 1 of year 1. Machine learning techniques, such as neural networks, could also model the data.

I am still waiting on access to the previously mentioned data sets due to the legal agreements and application process. In the meantime, I am working with synthetic telematics data from a published paper, " Synthetic Dataset Generation of Driver Telematics " . This dataset contains traditional variables used in insurance pricing, as well as telematics data, and claim numbers and claim amounts information for each observation. The data was generated to emulate a portfolio acquired from a Canadian-based insurer, which offered a UBI program.

Section C

I. RECORD OF CREDIT-BEARING MODULES

Date	Module Code and Title	Credits
28/09/2021 & 31/01/2022	STAT50080 - Foundations of Data Science I, STAT50090 - Foundations of Data Science II	5, 5
30/11/2021	ACM40660 - Scientific Programming (ICHEC)	5
19/04/2022	ACM40460 - High Performance Computing (ICHEC)	5



Section C - cont.

I. RECORD OF CREDIT-BEARING MODULES - cont.

Date	Module Code and Title	Credits
08/04/2022	RM8012 - Research Impact: Bridging the gap between rigour and relevance (University of Limerick)	6
18/04/2022	MP8002 - Entrepreneurial, creative and innovative thinking for researchers (University of Limerick)	6 aaa 6

II. RECORD OF ANY RECOGNITION OF PRIOR LEARNING AWARDED

Date	Description	Credits (if applicable)



Section D

RECORD OF PROFESSIONAL & CAREER DEVELOPMENT MODULES / WORKSHOPS

Date	Module Code and Title	Credits
22/03/2022	Computational Models in Immunology with Prof. Grant Lythe	
23/02/2022 - 24/02/2022	Mathematics Knowledge Exchange Workshop	
13/10/2022 & 01/11/2022	Imposter Syndrome and Staying Well in Your PhD with Hugh Kearns	
14/11/2022 - 16/11/2022	Ark Presentation Training	

Section E

RECORD OF RESEARCH INTEGRITY TRAINING

Date	Description eg. module code or certificate	Credits (if applicable)
20/01/2022	Research Integrity - Comprehensive award in Research Integrity Training	



Section F

RECORD OF OTHER RELEVANT ACTIVITIES

For example, you can record attendance / presentations at conferences, research visits, research mobility, networking events and placements / internships here (no more than 500 words):

Central Bank of Ireland (CBI), Dublin. 12 Week Enterprise Alliance Placement. 04/07/2022 - 23/09/2022.

2023 Winter Symposium, UCD. Attendee at conference. 17/01/2023 - 19/01/2023.

2022 Winter Symposium, Online. Presentation at conference: " Statistical and Machine Learning Methods for High Dimensional Macroeconomic and Financial Data. " 13/01/2022 - 14/01/2022.

Data on the Lake, Windermere, UK. Attendee at conference. 25/04/2022 - 29/04/2022.

Young ISA - Collaborative Workflow in Data Science, University of Limerick. Attendee at conference. 07/10/2022.

APTS - Statistical Computing & Statistics Inference, Durham University, UK. Attendee at conference. 04/04/2022 - 08/04/2022.

APTS - Applied Stochastic Processes & Statistical Modelling, Online. Attendee at conference. 13/12/2021 - 17/12/2021.

Individual Project: " Statistical Models for Improved Insurance Pricing and Risk Assessment " . 17/01/2022 - 12/07/2022

Group Project: " Statistical and Machine Learning Methods for High Dimensional Macroeconomic and Financial Data " . 20/09/2021 - 13/01/2022



Part 2 - RESEARCH STUDIES PANEL MEETING REPORT

To be completed by the chair of the RSP

Section A

PROGRESS REPORT

Progress to date (no more than 400 words):

James continued to perform very well during this term, showing excellent progresses in T&L, Research and School Contribution. He prepared a high-standard professional presentation, clearly showing his independence at work, initiative, engagement, and critical reflection. The panel was really impressed by his talk and encouraged him to continue the PhD at this standard. He has successfully completed additional 10 credits, 32 credits in total (FDS1, FDS2, RM8012, MP8002, ACM40660, ACM40460). He completed the Research Integrity Course and attended several workshops as the Imposter Syndrome, Staying Well in Your PhD, Ark Presentation Training. He attended the 2023 UCD Winter Symposium and Collaborative Workflow in Data Science UL. He successfully completed the 12 week Enterprise Alliance Placement in Central Bank of Ireland (CBI). He proactively contributed to a project aiming to detect liquidity problems in financial market because of Covid-19 disruptions. The main scope was to develop an automated way of detecting liquidity issues with a bond portfolio via AI techniques. He applied his newfound knowledge to a large dataset, made of an average of over 148,000 daily updates on over 250 variables (continuous, discrete, nominal or ordinal). He used a clustering approach, in the form of Dynamic Time Warping K-means, followed by supervised learning (LSTM and ARIMA). He described well the different components of the project - dataset cleaning, features selection, data analysis, model selection - showing confidence and mastery of the topics. He also well presented the results obtained and his original contribution on the project. He contributed to School activities, marking assignments and invigilating UCD exams. In terms of T&L, he has been tutor in STAT20100 and in ACM30110. It is evident from his description of the activities that he deeply reflected on the class delivery to improve student learning and engagement. In terms of research, he is working on two complementary projects supervised by Dr Adrian O'Hagan: Statistical Models for Improved Insurance Pricing and Risk Assessment Using Telematics and Statistical Models for Improved Insurance Pricing and Risk Assessment Using Wearables. James has already done an extensive literature review on both problems, and he is starting to work on available data. The panel was positively impressed by James and congratulated him.

Section B

Recommendations going forward (no more than 400 words):

The panel thinks that James has a clear plan for this year and for his future and we recommended that he continue in this way. The panel thinks he improved his presentation skills to present his work to a general audience. We strongly believe his experience at the Central Bank of Ireland has highly contributed to James' s professional growth.

The panel is enthusiastic of the research plan James has designed with his supervisor.

The first project addresses how Telematics information can be used to offer dynamic pricing, a.k.a Usage Based Insurance (UBI). He is going to analyzing telematics data provided by a global car rental company. This data set will contain information relating to vehicle maintenance, e.g. battery, oil, and fuel, as well as location information.

The second project is related to how health and activity data from wearable devices, e.g. smartwatches and fitness trackers, can be used to estimate claim likelihood and severity for health insurance policies. Wearable data come from the UK Biobank study. This longitudinal study has been running since 2006.

The first project is a continuation of his individual project, so he has already a clear plan in mind and expertise on analysis techniques. The panel advised him to start to work on synthetic data or open access dataset in the meantime he will receive the original dataset from the global car rental company to avoid delays in view of his STA exams.

We positively discovered that James is already working with synthetic telematics data from a published paper, " Synthetic Dataset Generation of Driver Telematics " as " back-up plan ". This dataset contains traditional variables used in insurance pricing, as well as telematics data.

For what concerns the second project, this is based on an active project started by Dr. Mark Farrell from Queen ' s University. An application to the UK Biobank exists titled, as well as a collaboration between Dr. Farrell and the supervisor. The panel is confident James will brilliantly contribute to this theme with his inventiveness and smartness. The panel also encouraged James to continue at this standard.



Section C

Additional comments (if any)

Section D

RESEARCH STUDIES PANEL MEETING RECORD REPORT

DECLARATION

We agree that this report is an accurate account of the meeting of the Doctoral Studies Panel.

Signature of PhD student James HANNON Date: 26/01/2023

Signature of Principal Supervisor Adrian O'Hagan Date: 23/02/2023

Signature of RSP Adviser 1 Adamaria Perrotta Date: 23/02/2023

Signature of RSP Adviser 2 Alessia Paccagnini Date: 23/02/2023

(Signature of RSP Adviser 3) Date:

Next meeting Date:

* Please sign or write your name in block capital letters

** Please upload to your SISWeb account and email a copy to the School Office



UCD Graduate Studies
www.ucd.ie/graduatestudies